BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 051 · Sequence, Input and Output

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Name one operating-system function.

Q2. What does software copyright restrict?
A. Using any keyboard
B. Learning to code
C. Copying without permission
D. Saving your own work

Q3. What is a typical feature of proprietary software?

Q4. A school needs a large, cheap archive drive. A pupil's only coursework copy is on an SSD. Correct both claims. Explain why. (a) An HDD is always faster and tougher than an SSD. (b) An SSD never fails, so needs no backup.

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I Do · Read new knowledge

Key vocabulary

Sequence — Instructions execute in order, so each value is ready before it is used.

Input — The input() command obtains data from a user and can store it in a variable.

Processing — Expressions calculate a new value from existing values.

Output — print() displays the result of a completed calculation.

Algorithm — An ordered set of instructions to solve a problem.

Trace — Following instructions and recording changes to values or output.

New learning

What is Sequence?: Sequence — Instructions execute in order, so each value is ready before it is used.

In this example: A bus ticket price app uses input; The input() command obtains data from a user and can store it in a variable.

Follow the effect: Expressions calculate a new value from existing values. Explain how this changes the eventual program result.

Check understanding: Output is used for a specific reason: print() displays the result of a completed calculation.

Worked example

A bus ticket price app is being written. Explain how sequence and input are used to make the program work. [4 marks]

1. We are designing a bus ticket price app. Identify which data and decisions are essential.

2. Sequence [1] — Instructions execute in order, so each value is ready before it is used. [1].

3. Input [1] — The input() command obtains data from a user and can store it in a variable. [1].

4. Two linked points for each technique give four mark points.

We Do · Practise with guidance

Guided 1

In a school uniform total, explain how sequence helps the program. [2 marks]

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Worked method: Sequence [1] — Instructions execute in order, so each value is ready before it is used. [1].

Guided 2

For a two sports scores, explain why input matters. [2 marks]

Worked method: Input [1] — The input() command obtains data from a user and can store it in a variable. [1].

Guided 3

When designing a cinema booking, explain how processing and output should be used. [4 marks]

Worked method: Processing [1] — Expressions calculate a new value from existing values. [1]. Output [1] — print() displays the result of a completed calculation. [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write one line to display the name the user enters. Assume the input is "Ava".

```
name = input("Your name?")
```

write

Question 1

State two programming terms or techniques relevant to this lesson. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

A shop receipt needs to work correctly. Explain how input and processing would help. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A pupil is programming a online café bill. Explain the roles of sequence, processing and output in their solution. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Instructions execute in order, so each value is ready before it is used.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Memory management, user management, file management, a user interface or peripheral management.

Q2. C — Copying without permission

Q3. Source code is normally controlled by its owner and users receive specified usage rights.

Q4. (a) An HDD can give lots of space at low cost per GB. Check the device and task. (b) An SSD can fail through faults or wear, so keep backups. No device is best at everything.

I Do / We Do pages · Guided practice

Q1. Sequence [1] — Instructions execute in order, so each value is ready before it is used. [1].

Q2. Input [1] — The input() command obtains data from a user and can store it in a variable. [1].

Q3. Processing [1] — Expressions calculate a new value from existing values. [1]. Output [1] — print() displays the result of a completed calculation. [1].

You Do pages · Code First

```
name = input("Your name?")
print(name)
```

Expected result: Ava (assuming input Ava)

You Do pages · Independent questions

Q1. Sequence [1]; Processing [1].

Q2. Input [1] — The input() command obtains data from a user and can store it in a variable. [1]. Processing [1] — Expressions calculate a new value from existing values. [1].

Q3. Sequence [1] — Instructions execute in order, so each value is ready before it is used. [1]. Processing [1] — Expressions calculate a new value from existing values.

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[1]. Output [1] — print() displays the result of a completed calculation. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: